Zeal Education Society's

ZEAL COLLEGE OF ENGINEERING & RESEARCH, PUNE – 41

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

NBA Accredited, NAAC Accredited with A+ Grade, ISO 21001:2018

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Curriculum Structure and Syllabus of

S.Y. B. Tech. – Electronics and Computer Engineering

(With effect from - Academic Year 2025 - 26)

(2024 Pattern)

VISION OF THE INSTITUTE

To be a premier institute in technical education by imparting academic excellence, research, social and entrepreneurial attitude.

MISSION OF THE INSTITUTE

- To achieve academic excellence through innovative teaching and learning process.
- To imbibe the research culture for addressing industry and societal needs.
- To inculcate social attitude through community engagement initiatives.
- To provide conducive environment for building the entrepreneurial skills.

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

VISION:

To emerge as a leading department in Electronics and Computer Engineering by imparting academic excellence, fostering innovation and research, and nurturing socially responsible and entrepreneurial professionals for sustainable technological growth.

MISSION:

M1: To facilitate quality education in Electronics and Computer Engineering through innovative pedagogy and hands-on learning.
M2: To foster research and development addressing real-world industrial and societal challenges.
M3: To instill ethical values and social responsibility through active community engagement.
M4: To create a conducive ecosystem for developing entrepreneurial skills and leadership Qualities using Industry-Academia Collaboration.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs):

PEO1: Graduates will establish themselves as professionals by applying core concepts of electronics and computer engineering to solve real-world problems.
PEO2: Graduates will adapt to emerging technologies through lifelong learning and pursue higher education, certifications, or entrepreneurship.
PEO3: Graduates will demonstrate ethical values, communication skills, and leadership qualities while working in multidisciplinary teams.

PROGRAM OUTCOMES (POs):

PO1: **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2: **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3: **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs):

PSO1: Apply principles of electronics, signal processing, and embedded systems to develop hardware/software-based solutions.

PSO2: Design and implement computer engineering solutions involving databases, networking, and algorithms for real-time and data-centric applications.

PSO3: Integrate domain knowledge with modern tools and technologies to solve interdisciplinary engineering problems in IoT, automation, and intelligent systems.

LIST OF ABBREVIATIONS

Abbreviation	Description
BSC	Basic Science Course
ESC	Engineering Science Course
PCC	Program Core Course
PEC	Program Elective Course
MDM	Multidisciplinary Minor
OE	Open Elective - Other than a particular program
VSEC	Vocational and Skill Enhancement Course
AEC	Ability Enhancement Course
ENTR	Entrepreneurship
EC	Economics
MC	Management Courses
IKS	Indian Knowledge System
VEC	Value Education Courses
RM	Research Methodology
CEP	Community Engagement Project
FP	Field Project
PROJ	Project
INT	Internship
OJT	On Job Training
CC	Co-curricular Courses
HSSM	Humanities Social Science and Management
ELC	Experiential Learning Course
B. Tech	Bachelor of Technology
L	Lecture
P	Practical
T	Tutorial
H	Hours
CR	Credits
CIE	Continuous Internal Evaluation
ETE	End Term Evaluation
TW	Term Work
OR	Oral
PR	Project

Second Year B. Tech. – Electronics and Computer Engineering: Semester - III

Course Code	Course Type	Course Name	Teaching Scheme (hrs/Week)							Evaluation Scheme					
			L	P	T	H	CR			CIE	ETE	TW	PR	OR	Total
							TH	PR/Tut	Total						
ECPC302	PCC	Digital Systems Design & Applications	3	2	-	5	3	1	4	40	60	-	50	-	150
ECPC303	PCC	Data Structure and Algorithms	3	2	-	5	3	1	4	40	60	-	-	25	125
ECPC304	PCC	Computer Organization and Architecture	3	-	-	3	3	-	3	40	60	-	-	-	100
ECMD301	MDM	Engineering Mathematics III	3	-	-	3	3	-	3	40	60	-	-	-	100
ALOE301	OE	Open Elective – I#	2	-	-	2	2	-	2	40	60	-	-	-	100
ECMC301	HSSM-MC	Project Management System – I	-	2	-	2	-	1	1	-	-	25	-	-	25
ECVS303	VSEC	Problem Solving Technique – I	-	2	-	2	-	1	1	-	-	25	-	-	25
ECCE301	CEP	Project Based Learning	-	2	-	2	-	1	1	-	-	25	-	-	25
ECVS304	VSEC	Electronics Workshop	-	2	-	2	-	1	1	-	-	25	-	-	25
ECIN302	ELC - INT	Internship – II	4 Weeks				-	2	2	-	-	25	-	-	25
Total			**14**	**12**	**-**	**26**	**14**	**8**	**22**	**200**	**300**	**125**	**50**	**25**	**700**

# - Select any one course from the given Open Elective Courses		
Course Code	**Course Type**	**Open Elective - I**
ALOE301A	OEC	Digital Literacy and Applications
ALOE301B		Environmental Studies
ALOE301C		Green Energy and Sustainability
ALOE301D		Basics of Consumer Electronics
ALOE301E		Renewable Energy Systems

BoS Chairman

PU/PN/Engg./285/2007
ZES's Zeal College of Engineering & Research * Narhe, Pune-41 *

Director
ZES's Zeal College of Engineering & Research Narhe, Pune - 411041.

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Second Year B. Tech. – Electronics and Computer Engineering: Semester - IV

Course Code	Course Type	Course Name	Teaching Scheme (hrs/Week)							Evaluation Scheme					
			L	P	T	H	CR			CIE	ETE	TW	PR	OR	Total
							TH	PR/Tut	Total						
ECPC405	PCC	Database Management	3	2	-	5	3	1	4	40	60	-	-	25	125
ECPC406	PCC	Communication Systems	3	2	-	5	3	1	4	40	60	-	-	25	125
ECPC407	PCC	Object Oriented Programming	3	2	-	5	3	1	4	40	60	-	25	-	125
ECMD402	MDM	Sensors and Applications	3	-	-	3	3	-	3	40	60	-	-	-	100
ALOE402	OE	Open Elective – II#	2	-	-	2	2	-	2	40	60	-	-	-	100
ECMC402	HSSM-MC	Quality Management System – II	-	2	-	2	-	1	1	-	-	25	-	-	25
ECAE402	AEC	Problem Solving Technique – II	-	2	-	2	-	1	1	-	-	25	-	-	25
ECVS405	VSEC	Python Programming	-	2	-	2	-	1	1	-	-	-	-	50	50
ECIN403	ELC - INT	Internship – III	4 Weeks				-	2	2	-	-	25	-	-	25
Total			**14**	**12**	**-**	**26**	**14**	**08**	**22**	**200**	**300**	**75**	**25**	**100**	**700**

# - Select any one course from the given Open Elective Courses		
Course Code	**Course Type**	**Open Elective - II**
ALOE402A	OEC	Cyber Security and Laws
ALOE402B		Sustainability and Climate Change
ALOE402C		Energy Audit and Electrical Safety
ALOE402D		Digital Marketing
ALOE402E		Entrepreneurship and Innovations

Mahesh
BoS Chairman

PU/PN/Engg./285/2007
ZES's Zeal College of Engineering & Research
Narhe, Pune-41

Director
ZES's Zeal College of Engineering & Research
Narhe, Pune - 411041.

INDEX

SYLLABUS

SEMESTER - III

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Program: B. Tech. (Electronics and Computer Engineering)								**Semester:** III	
Course: Digital Systems Design & Applications								**Code:** ECPC302	
Teaching Scheme (Hrs/week)				**Evaluation Scheme (Marks)**					
Lecture	**Practical**	**Tutorial**	**Credit**	**CIE**	**ETE**	**TW**	**OR**	**PR**	**Total**
03	02	-	04	40	60	-	-	50	150

Prerequisites:

Number systems (binary, decimal), basic concepts of electricity and circuits, Boolean algebra for logic operations, and fundamental knowledge of electronic components.

Course Objectives:

1. To provide foundational knowledge in combinational and sequential digital logic circuit design.
2. To enable students to analyze, design, and implement state machines and programmable logic devices.
3. To introduce modern digital design techniques using PLDs, FPGAs, and hardware description languages like Verilog.

Course Outcomes: After completion of this course, students will able to -

CO1	Design and implement combinational circuits using digital components.
CO2	Explain flip-flops, shift registers, and counters in circuits.
CO3	Analyze and design FSMs, implement sequence detectors.
CO4	Design circuits using ROM, PLA, PAL and understand applications.
CO5	Apply digital logic using ROMs, CPLDs, and FPGAs.
CO6	Gain VLSI knowledge, understand Verilog and FPGA architecture.

Course Contents:

Unit	Description	Duration (Hrs.)
1.	**Combinational Circuit Design:** Digital Codes: Binary, BCD, Grey, Excess-3. Code Conversions: Binary to Grey, BCD to Excess-3 and its applications. Half and Full Adder, Half & Full Subtractor, Digital Comparator, Digital Comparator with multiple inputs, Realization of Boolean functions using Multiplexer/ demultiplexer, Parity generator and checker (Even & Odd).	07
2.	**Sequential Circuit Design:** Flip-Flops: SR, JK, D, T flip-flops, Preset & Clear operations, Truth Tables and Excitation Tables. Conversion of flip flops, Typical data sheet specifications of Flip flop, application of Flip flops. Registers: Buffer registers, Shift registers (SISO, SIPO, PISO, PIPO) Counters: Asynchronous and Synchronous Counters, Ring counter, Johnson counter, Modulus counter (IC 7490), Pulse train generator.	07

3.	**State Machines:** Basic design steps- State diagram, State table, State reduction, State assignment, Mealy and Moore machines representation, Implementation, finite state machine implementation, Sequence detector. Introduction to Algorithmic state machines- construction of ASM chart and realization for sequential circuit.	07
4.	**Programmable Logic Devices:** Programmable Logic Devices (PLD): Introduction to PLDs: ROM, PLA, PAL, Designing Combinational Circuits using PLDs, Applications of PLDs in digital circuit design.	07
5.	**Applications of Digital Circuits:** Introduction to Digital Circuits - Design of Sequence Detector, Design of Iterative circuit (Comparator), Design of sequential circuits using ROM & PLAs, CPLDs & FPGAs, Serial adder with Accumulator.	07
6.	**Introduction VLSI:** Introduction to VLSI, Introduction to Hardware description languages (Verilog), Verilog Concepts, Basic concepts-Modules & ports & Functions, useful modeling techniques, Introduction to FPGA Architecture.	07
	TOTAL	**42**

List of Experiments:

Perform a total of 8 experiments out of the 12 listed below:

- **Select any 6 experiments from Group A**
- **Select any 2 experiments from Group B**

Group A

1. Design and implement code converters- Binary to Gray and BCD to Excess-3
2. Design and implement of Half Adder/ Full Adder using a) Basic Gates b) Universal Gates
3. Realization of Boolean function using Multiplexer 74151/74153, Demultiplexer 74154 / 74138.
4. Design and implementation of 1-bit comparator and 2-bit comparator
5. Design and implementation of parity generator
6. Verify characteristic tables of SR, JK, D & T Flip-flop
7. Design and implementation of Asynchronous/synchronous 3-bit counter using D flip-flop
8. Design and implement of Sequence generator/ detector using JK flip-flop
9. Design and implement MOD-10 counter using IC7490

Group B

1. Implement a digital circuit using FPGA (Blinking LED using Simple Timer Circuit or 4-bit Binary Counter)
2. Building Combinatorial Circuit Using Data Flow Modeling Lab (https://download.ni.com/pub/gdc/tut/dataflow_lab.pdf)
3. Study modeling techniques for efficient circuit design in Verilog.

Text Books :
1. M. Morris Mano, Michael D. Ciletti, "Digital Logic and Computer Design" , Pearson Education / Prentice Hall 2. R. P. Jain , "Modern Digital Electronics" , McGraw Hill Education
Reference Books:
1. Donald P. Leach, Albert Paul Malvino, and Goutam Saha, "Digital Principles and Applications", Tata McGraw-Hill. 2. Ramesh Gaonkar, "Microprocessor Architecture, Programming, and Applications with the 8085", Pearson Prentice Hall. 3. Muhammad Ali Mazidi, Janice Mazidi, and Rolin McKinlay, "The 8051 Microcontroller and Embedded Systems", Pearson India / Prentice Hall PTR. 4. M. Morris Mano and Michael D. Ciletti, "Digital Design: With an Introduction to the Verilog HDL", Pearson / Pearson Prentice Hall.
E-Resources:
1. **Combinational Circuit Design,** [Neso Academy – Combinational Logic Playlist (YouTube)] https://www.youtube.com/playlist?list=PLBlnK6fEyqRhX6r2uhhlubuF5QextdCSM 2. **Sequential Circuit Design,** [NPTEL Course: Digital Circuits – IIT Madras] https://nptel.ac.in/courses/117/106/117106086/ 3. **FSM and ASM Design,** [GeeksforGeeks – Finite State Machines Explained] https://www.geeksforgeeks.org/finite-state-machine-types-design-working-and-applications/ 4. **Programmable Logic Devices (PLDs),** [TutorialsPoint – Programmable Logic Devices (ROM, PLA, PAL)] https://www.tutorialspoint.com/digital_circuits/digital_circuits_programmable_logic_devices.htm 5. **Applications of Digital Circuits,** YouTube – Sequence Detector using Verilog 6. **Introduction to VLSI & Verilog,** HDLBits – Verilog Practice Problems (Beginner to Advanced)] https://hdlbits.01xz.net/wiki/Main_Page

Program: B. Tech. (Electronics and Computer Engineering)	**Semester:** III
Course: Data Structures and Algorithms	**Code:** ECPC303

Teaching Scheme (Hrs/week)				**Evaluation Scheme (Marks)**					
Lecture	**Practical**	**Tutorial**	**Credit**	**CIE**	**ETE**	**TW**	**OR**	**PR**	**Total**
03	02	-	04	40	60	-	25	-	125

Prerequisites: Fundamental knowledge of C programming.

Course Objectives:

1. To introduce fundamental concepts, types, and operations of data structures with attention to time and space complexity.
2. To develop the ability to design and implement various data structures like arrays, linked lists, stacks, queues, trees, and graphs.
3. To enhance problem-solving skills using efficient algorithms for searching, sorting, and graph traversal techniques

Course Outcomes: After completion of this course, students will be able to -

CO1	Explain data structures, classifications, operations, and analyze algorithm complexity.
CO2	Compare sorting and searching algorithms based on time complexity.
CO3	Design and perform operations on arrays and linked lists.
CO4	Implement stack, queue using arrays, lists in applications.
CO5	Construct and manipulate binary and AVL trees with traversals.
CO6	Apply graph techniques and algorithms for graph-based problems.

Course Contents:

Unit	Description	Duration (Hrs.)
1.	**Introduction to Data Structure:** Concept, Types of data structures, Common operations on data structures, Complexities, Time Complexity, order of Growth, Asymptotic Notation.	07
2.	**Sorting and Searching Techniques:** Sorting and Searching techniques: Introduction, Sorting, Insertion Sort, Selection Sort, Bubble Sort, Merge-Sort, Linear search and Binary Search.	07
3.	**Linear Array and Linked List:** **Linear Arrays:** Introduction, Linear Arrays, Representation of Linear array in Memory, Traversing Linear Arrays, Insertion and deletion, 2D & Multi-dimensional Array, Sparse matrix. **Linked List:** Introduction to Linked Lists, Representation of Linked Lists in Memory, Traversing a Linked List, Searching a Linked List, Insertion into a Linked List, Deletion from a Linked List, Circularly Linked Lists, Doubly Linked Lists.	07

4.	**Stacks, and Queues in Data Structures:** **Stacks:** Introduction, Stacks, Array Representation of Stacks, Linked Representation of Stacks, Arithmetic Expression; Polish Notation, Recursion, Towers of Hanoi, **Queue:** Introduction, Linked Representation of Queues, Circular Queues, De-queue, Priority Queues	07
5.	**Trees in Data Structures:** **Trees:** Basic terminology. Binary Tree: Properties of a Binary Tree, ADT Binary trees and its representations. **Operations:** Insert, Delete & **Traversal:** Preorder, In order, Post order, **Binary Search Trees:** Searching and Inserting in Binary Search Trees, Deleting in a Binary Search Tree, Balanced Binary Trees, **AVL Search Trees:** Insertion in an AVL Search Tree, Deletion in an AVL Search Tree.	07
6.	**Graphs Theory in Data Structures:** **Graphs:** Introduction to graphs, Graph Theory Terminology, Sequential Representation of Graphs, Adjacency Matrix; Path Matrix, Linked Representation of a Graph, Operations on Graphs, Traversing a Graph, BFS and DFS, Spanning Trees, Minimum Spanning Trees Kruskal's and Prim's algorithm, Dijkstra's algorithm.	07
	TOTAL	**42**

List of Experiments:

Perform any 08 experiment out of 10:

1. Write a C program to implement a linear search and Binary Search for a given array.
2. Write a C program to arrange the list of students according to roll numbers in ascending order using 1) Bubble Sort 2) Insertion sort
3. Write a C program to implement a sparse matrix with operations like initialize empty sparse matrix, insert an element, sort a sparse matrix on row-column, transpose a matrix, etc.
4. Write a C program to develop a hash table to implement hashing. **(Content Beyond Syllabus)**
5. Write a C program to write functions to 1) Add and delete the nodes in a linked list. 2) Compute total number of nodes in the linked list 3) Display list in reverse order using recursion.
6. Write a C program to implement stack using a linked list and perform evaluation of a postfix expression using stack.
7. Write a C program to implement queue operations.
8. Write a C program to implement tower of hanoi using recursion.
9. Write a C program to implement tree traversal.
10. Write a C program to implement graph traversal.
11. **Perform at least one practical using virtual lab (VLab). (Compulsory)**

Text Books:
1. Seymour Lipschutz, "Schaum's Outline of Data Structures", McGraw-Hill Companies Incorporated. 2. Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman, "Data Structures and Algorithms", Addison-Wesley, 1983. 3. Ellis Horowitz, Sartaj Sahni, Susan Anderson-Freed, "Fundamentals of Data Structures in C", University Press, Second Edition, 2008.
Reference Books:
1. Alfred V. Aho, Jeffrey D. Ullman, "Data Structures & Algorithms", 1st Edition, Pearson. 2. Michael T. Goodrich, Roberto Tamassia, David M. Mount, "Data Structures and Algorithms in Java", 5th Edition, Wiley, 2010.
E-Resources:
NPTEL Course: 1. https://nptel.ac.in/courses/106/102/106102064/ 2. http://cse01-iiith.vlabs.ac.in/ 3. https://ds2-iiith.vlabs.ac.in/data-structures-2/

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Program: B. Tech. (Electronics and Computer Engineering)								**Semester:** III	
Course: Computer Organization and Architecture								**Code:** ECPC304	
Teaching Scheme (Hrs/week)				**Evaluation Scheme (Marks)**					
Lecture	**Practical**	**Tutorial**	**Credit**	**CIE**	**ETE**	**TW**	**OR**	**PR**	**Total**
03	-	-	03	40	60	-	-	-	100

Prerequisites: Basics of computer systems

Course Objectives:

1. Understand computer evolution, components, interconnections, buses, and apply binary arithmetic on signed and unsigned numbers.
2. Understand CPU operations and compare RISC and CISC architectures.
3. Analyze hardwired control unit design and control unit functions.
4. Design memory systems considering trade-offs and performance.
5. Analyze pipelining for efficient instruction execution with minimal hazards.
6. Explain I/O transfer techniques and the operation of various I/O peripherals.

Course Outcomes: After completion of this course, students will able to -

CO1	Understand the structure, function, and characteristics of computer systems.
CO2	Describe the function of the Central Processing Unit and RISC and CISC Architecture.
CO3	Explore the knowledge about Control Unit Design.
CO4	Analyze trade-offs and performance issues.
CO5	Apply a pipeline for consistent execution of instructions.
CO6	Discuss the working mechanisms of various I/O peripherals

Course Contents:

Unit	Description	Duration (Hrs.)
1.	**Computer Evolution & Arithmetic:** Organization & Architecture, Structure & Function, Integer Representation: Fixed point & Signed numbers. Integer Arithmetic: 2's Complement arithmetic, multiplication, Booth's Algorithm, Division Restoring Algorithm, Non-Restoring algorithm.	07
2.	**Processor Design:** CPU Architecture, Register Organization, Instruction types, Types of operands, Types of operation, Instruction formats, Addressing modes and address translation. Instruction cycles. RISC Processors: RISC- Features, CISC Features, Comparison of RISC & CISC Processors. Comparison between Harvard and Von Neumann Architecture Case study of Processor: Von Neumann Architecture.	07

3.	**Control Unit:** Fundamental Concepts: Single Bus CPU organization, register transfers, performing an arithmetic/ logic operation, fetching a word from memory, storing a word in memory, Execution of a complete instruction. Micro-operations, Types of Control Unit, Hardwired Control, Micro-programmed Control: Microinstructions.	07
4.	**Memory Organization:** Need, Hierarchical memory system, Characteristics, Size, Access time, Read Cycle time and address space. Main Memory Organization: ROM, RAM, EPROM, E2PROM, DRAM, Cache memory Organization, Cache Mapping techniques: Direct, Set Associative, Fully Associative.	07
5.	**Pipelining:** Data hazards: operand forwarding, handling data hazards in software, side effects. Instruction hazards: unconditional branches, conditional branches, and branch prediction. Performance considerations: effect of instruction hazards, number of pipeline stages.	07
6.	**I/O Organization:** Input/output systems, I/O Transfer Techniques: Program-controlled, Interrupt-Driven, DMA controlled synchronous, asynchronous, working mechanisms of peripherals: keyboard, video displays, touch screen panel, printers	07
	TOTAL	**42**

Text Books :

1. William Stallings, "Computer Organization and Architecture: Designing for Performance", 7th Edition, Pearson Prentice Hall Publication.
2. C. Hamacher, V. Zvonko, S. Zaky, "Computer Organization", 5th Edition, Tata McGraw Hill Publication.
3. Kai Hwang, "Advanced Computer Architecture", Tata McGraw-Hill.

Reference Books:

1. Hwang and Briggs, "Computer Architecture and Parallel Processing", Tata McGraw Hill Publication.
2. A. Tanenbaum, "Structured Computer Organization", Prentice Hall Publication, 4th Edition.

E-Resources:

1. www.nptelvideos.in
2. www.geeksforgeeks.org
3. www.udemy.com

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Program: B. Tech. (Electronics and Computer Engineering)								**Semester:** III	
Course: Engineering Mathematics – III								**Code:** ECMD301	
Teaching Scheme (Hrs/week)				**Evaluation Scheme (Marks)**					
Lecture	**Practical**	**Tutorial**	**Credit**	**CIE**	**ETE**	**TW**	**OR**	**PR**	**Total**
03	-	-	03	40	60	-	-	-	100

Prerequisites:

First order and first degree differential equations, calculus and vector differentiation.

Course Objectives:

1. To familiarize students with higher-order differential equations, transforms, statistics, probability, and vector calculus concepts.
2. To equip students with mathematical techniques to enhance analytical thinking and solve discipline-specific problems.

Course Outcomes: After completion of this course, students will able to -

CO1	Solve higher-order differential equations and model electrical circuits.
CO2	Analyze data using statistical and probability concepts.
CO3	Apply Z-transform concepts in digital signal processing.
CO4	Understand Laplace transform and use it in applications.
CO5	Evaluate Fourier transforms and apply in signal processing.
CO6	Apply vector integral calculus in electromagnetic field problems.

Course Contents:

Unit	Description	Duration (Hrs.)
1.	**Linear Differential Equations:** Linear Differential Equations (LDE) of nth order with constant coefficients, Method of variation of parameters, Cauchy's and Legendre's D.E., Simultaneous DE and applications of differential equations to electric circuits.	07
2.	**Statistics and Probability:** Measures of central tendency, Measures of dispersion, Moments, Skewness and Kurtosis, Correlation and Regression. Definition and theorems on Probability, Probability Distributions: Binomial distribution Poisson distribution, Normal distribution, Test of Hypothesis: Chi-Square test.	07
3.	**Z- Transforms:** Definition, Properties of Z-transform, Z- transform of Standard Sequences. Inverse, Z-transform, Solution of difference equation by Z-transform.	07
4.	**Laplace Transform:** Definition and properties of Laplace transform, Inverse Laplace transform,	07

	Applications of Laplace transform to solve differential equation.	
5.	**Fourier Transforms:** Fourier Transform (FT): Complex Exponential Form of Fourier Series, Fourier Transform, Inverse Fourier Transform, Fourier Sine transform, Fourier Cosine transform, Inverse Fourier Sine Transform, Inverse Fourier Cosine Transform.	07
6.	**Vector Integral Calculus & Applications:** Line integral, Work done, Green's Lemma, Gauss's Divergence Theorem, Stroke's theorem. Applications of vector integral calculus in Electro-magnetic field.	07
	TOTAL	**42**

Text Books :

1. B.S. Grewal, "Higher Engineering Mathematics", 40^{th} Edition, Khanna Publishers, Delhi, 2008.
2. P. N. Wartikar, J.N. Wartikar, "Applied Mathematics, Volumes I and II", Pune Vidyarthi Griha Prakashan, Pune.
3. H.K. Das, "Higher Engineering Mathematics", S. Chand Publication.

Reference Books:

1. Erwin Kreyszig, "Advanced Engineering Mathematics", 10th Edition, Wiley Publications, 2015.
2. Sheldon M. Ross, "Introduction to Probability and Statistics for Engineers and Scientists", 5^{th} Edition, Elsevier Academic Press.
3. B.V. Raman, "Engineering Mathematics", Tata McGraw-Hill.
4. C.R. Wylie, L.C. Barrett, "Advanced Engineering Mathematics", McGraw-Hill, Inc.
5. Thomas L. Harman, James Dabney, Norman Richert, "Advanced Engineering Mathematics with MATLAB", 2nd Edition, Brooks/Cole, Thomson Learning.
6. Joel Hass, Christopher Heil, "Calculus", 11th Edition, Pearson, 2016. (Assumed based on partial information—please confirm if it's "Calculus, 11th Edition").

E-Resources:

1. NPTEL Course "Transform Calculus and its Applications in Differential Equations"
   https://nptel.ac.in/courses/111/105/111105123/
2. NPTEL Course "Probability Theory and Applications"
   https://nptel.ac.in/courses/111/104/111104079/

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Program: B. Tech. (Electronics and Computer Engineering)							**Semester:** III		
Course: Project Management System – I							**Code:** ECMC301		
Teaching Scheme (Hrs/week)				**Evaluation Scheme (Marks)**					
Lecture	**Practical**	**Tutorial**	**Credit**	**CIE**	**ETE**	**TW**	**OR**	**PR**	**Total**
-	02	-	01	-	-	25	-	-	25

Prerequisites:

Interactive mind-set for practical.

Course Objectives:

1. To acquire basic knowledge of Problem-solving techniques.
2. To understand the structured way of solving problems with the right tools.

Course Outcomes: After completion of this course, students will be able to -

CO1	Know the project and its importance.
CO2	Understand the structured way of project execution process.
CO3	Understand on how to project, goals and timeline.
CO4	Know the key principles of project management.

Course Contents:

Unit	Description	Duration (Hrs.)
1.	**Project & Management System:** What is a project, What is Project Management, Types, Importance and its benefits.	06
2.	**Project Management Process:** Planning, Execution, Monitoring & Control, Deliverables, Stakeholders.	06
3.	**Principles:** 12 Principles of Project Management.	16
	TOTAL	**28**

Text Books:

1. K. Nagarajan, "Project Management", New Age International Publishers.
2. Joseph Heagney**,** "Fundamentals of Project Management", AMACOM.
3. Harold Kerzner**,** "Project Management: A Systems Approach to Planning, Scheduling, and Controlling", Wiley.

Reference Books:

1. "A Guide to the Project Management Body of Knowledge (PMBOK Guide)", Project Management Institute.
2. BB Goel, "Project Management: Principles and Techniques", Deep & Deep Publications Pvt. Ltd.

E-Resources:

1. Dr. Nimisha Singh, "Introduction to Project Management: Principles & Practices", NPTEL Course - https://onlinecourses.swayam2.ac.in/imb25_mg167/preview
2. Prof. Raghu Nandan Sengupta, "Project Management", NPTEL Course - https://onlinecourses.nptel.ac.in/noc25_mg78/preview

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Program: B. Tech. (Electronics and Computer Engineering)								**Semester:** III	
Course: Problem Solving Techniques – I								**Code:** ECVS304	
Teaching Scheme (Hrs/week)				**Evaluation Scheme (Marks)**					
Lecture	**Practical**	**Tutorial**	**Credit**	**CIE**	**ETE**	**TW**	**OR**	**PR**	**Total**
-	02	-	01	-	-	25	-	-	25

Prerequisites:

Interactive mind-set for practical.

Course Objectives:

1. To acquire basic knowledge of Problem-solving techniques.
2. To understand the structured way of solving problems with the right tools.

Course Outcomes: After completion of this course, students will be able to -

CO1	Know the problem and types of problem.
CO2	Understand the structured way of solving a problem.
CO3	Understand the basic tools and its application.
CO4	Apply the learning to solve simple problem cases as a team.

Course Contents:

Unit	Description	Duration (Hrs.)
1.	**Problem Understanding:** Define problem, Types of Problem, What is problem solving? Structured way of Problem solving – Step by Step.	06
2.	**Problem Solving Approach:** Structured step by step working model, Principles to think and apply.	06
3.	**Basic Tools for Problem Solving:** Knowing the tools and applying the right tools at the right step of problem solving, Problem solving case study.	16
	TOTAL	**28**

Text Books:

1. M.T. Somashekara, "Problem Solving and Programming Concepts", PHI Learning.
2. Dheeraj Sharma, "Problem Solving and Decision Making", McGraw-Hill Education.

Reference Books:

1. Willian Henderson, "Master Critical Thinking, Creative, Logic & Problem solving skills", Peak Publish LLC.
2. Sharma Narender, "Handbook 7 QC tools", Shakehand with Life.

E-Resources:

1. Coursera: "Creative Problem Solving" - https://www.coursera.org/learn/creative-problem-solving.
2. MindTools – "Problem Solving Techniques", https://www.mindtools.com/cx4ems0/problem-solving.

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Program: B. Tech. (Electronics and Computer Engineering)								**Semester:** III	
Course: Project Based Learning								**Code:** ECCE301	
Teaching Scheme (Hrs/week)				**Evaluation Scheme (Marks)**					
Lecture	**Practical**	**Tutorial**	**Credit**	**CIE**	**ETE**	**TW**	**OR**	**PR**	**Total**
-	02	-	01	-	-	25	-	-	25

Prerequisites:

Basic idea of report writing, fundamental knowledge of electronics and software languages is required

Course Objectives:

1. To highlight long-term, multidisciplinary, and student-centered project-based learning activities.
2. To foster both individual and group learning by using the tools at hand to solve real-world problems.
3. To be able to create applications based on the principles of telecommunication and electronics engineering, sometimes using previously learned material.
4. To gain hands-on experience in all phases of the electrical system development life cycle, including design, implementation, testing, and specification.
5. To be able to develop and assess the suggested system using the right hardware and software tools.
6. To provide each student the chance to participate, either alone or in a group, in orders to foster professionalism and teamwork.

Course Outcomes: After completion of this course, students will able to -

CO1	Recognize a significant real-world issue, potentially involving multiple disciplines, by conducting a comprehensive literature review, and design appropriate aims and objectives to guide the project.
CO2	Use of ethical practices and safety standards consistently while implementing a solution that offers meaningful societal benefits.
CO3	Develop an innovative solution rooted in electronics and telecommunication engineering by combining foundational concepts with existing knowledge and practical experience.
CO4	Implement appropriate technologies in the project and present your understanding and learning through both verbal explanations and written reports.
CO5	Develop both independent working skills and effective teamwork capabilities to contribute meaningfully in various collaborative settings.
CO6	Differentiate between individual and group roles by examining responsibilities in collaborative tasks, and organize personal contributions to enhance overall team performance.

Group Structure:

Working in supervisor/mentor –monitored groups. The students plan, manage and complete a task/project/activity which addresses the stated problem.

1. **Create groups of 5 (five) to 6 (six) students in each class**.

Project Selection:

Analyse the problem, design, and determine the values of the components. Survey through journals, patents, or field visits (a problem can be theoretical, practical, social, technical, symbolic, cultural, and/or scientific). There are no widely accepted standards for what makes a project acceptable. Projects differ substantially in terms of the activity's substance and structure, the clarity of the learning objectives, and the depth of the questions examined. It is advised to use the problem-based, project-oriented learning methodology. The concept starts with the identification of an issue, which frequently develops from a query or "wondering." The learning process then begins with this problem formulation.

A dilemma arises from students' curiosity in various academic fields and professional settings and might be theoretical, practical, social, technical, symbolic, cultural, and/or scientific. The project topic may be interdisciplinary in view, as indicated in the preamble, since electronics serves as a crucial foundation for other fields (computer science, signal processing, and communications). Although, the selected challenge needs to use the principles of electronics and telecommunication engineering. Electronic components must make up at least 40% of the project's overall established system setup. However, in an actual instance, a project topic that is entirely software-based might be permitted.

Effective Documentation:

Effective writing skills must be taught to students in order for our engineering graduates to be able to provide documentation that works. The literature review, problem statement, aim and objectives, system block diagram, system implementation details, discussion and analysis of the results, conclusion, system limitations, and future scope are all intended to be included in the DSP final report. The creation of the DSP synopsis and final report is anticipated to involve the usage of numerous publicly accessible software tools, such as Medley (Elsevier) and Grammerly. It is anticipated that DSP mentors and guides will instruct students on how to use reliable sources of knowledge on their DSP topic, including books, magazines, and reference papers.

Evaluation & Continuous Assessment:

The organization, leader, or mentor is dedicated to analysing and evaluating program efficacy as well as student success. Course progress is routinely tracked every week. The work needs to be reviewed once a week. Individual and team performance must be measured throughout the monitoring, ongoing assessment, and evaluation process. Authorities and supervisors/mentors oversee the course content and do ongoing evaluations. Students are required to uphold an institutional culture that values genuine teamwork, self-motivation, think, learn and share peer learning processes, and individual accountability. Through guidance and orientation programs, as well as the provision of suitable resources and services, the department or institution should assist students in this respect. Students and their supervisors/mentors must actively engage in the assessment and evaluation procedures. It is advised that all activities be routinely and legally documented.

Recommended parameters for assessment, evaluation and weightage:

1. Idea Inception (kind of survey). (10%)
2. Outcome (Participation/ publication, copyright, patent, product in market). (50%)
3. Documentation (Gathering requirements, design & modeling, implementation/execution, use of technology and final report, other documents). (15%)

4. Attended reviews, poster presentation and model exhibition. (10%)
5. Demonstration (Poster Presentation, Model Exhibition etc.) (10%).
6. Awareness /Consideration of - Environment/ Social /Ethics/ Safety measures/Legal aspects. (5%)

Reference Books and Research Articles :

1. John Larmer, John R. Mergendoller, and Suzie Boss, "Setting the Standard for Project Based Learning".
2. John Larmer and Suzie Boss, "Project Based Teaching: How to Create Rigorous and Engaging Learning Experiences".
3. Erin M. Murphy and Ross Cooper, "Hacking Project Based Learning: 10 Easy Steps to PBL and Inquiry".
4. M. Krašna, "Project based learning (PBL) in the teachers' education," 39th International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO), Opatija, 2016, pp. 852-856, doi: 10.1109/MIPRO.2016.7522258.
5. J. Macias- Guarasa, J.M. Montero, R. San-Segundo, A. Araujo and O. Nieto-Taladriz, "A project based learning approach to design electronic systems curricula", IEEE transactions on Education, vol.49, no. 3, pp. 389-397, Aug. 2006, doi: 10.1109/TE.2006.879784.

E-Resources:

1. Project-Based Learning, Edutopia, March 14, 2016. https://www.edutopia.org/project-based-learning
2. www.howstuffworks.com.
3. Condliffe, Barbara. "Project-Based Learning: A Literature Review. Working Paper." MDRC (2017). https://eric.ed.gov/?id=ED578933
4. Activity-Based Learning. https://genevaglobal.com/wp-content/uploads/2021/10/Activity-Based-Learning.GenevaGlobal.2021-07.pdf

NPTEL Resources:

1. Problem Based Learning by Dr. Indrajit Saha, National Institute of Technical Teachers Training and Research, Kolkata.

Program: B. Tech. (Electronics and Computer Engineering)							**Semester:** III		
Course: Electronics Workshop							**Code:** ECVS304		
Teaching Scheme (Hrs/week)				**Evaluation Scheme (Marks)**					
Lecture	**Practical**	**Tutorial**	**Credit**	**CIE**	**ETE**	**TW**	**OR**	**PR**	**Total**
-	02	-	02	-	-	25	-	-	25

Prerequisites: Basics Electronics and Electrical Engineering

Course Objectives:

1. Familiarize students with electronic simulator tools.
2. Enable practical understanding of error analysis, bridge circuits, and waveform measurements.
3. Introduce sensors and transducers used in automation systems.

Course Outcomes: After completion of this course, students will able to -

CO1	Apply basic MATLAB commands for matrix operations and plotting.
CO2	Implement fundamental electronic circuit components using Multisim.
CO3	Understand microcontroller architecture and basic programming.
CO4	Create and edit PCB layouts.
CO5	Develop and modify PCBs.

Course Contents:

List of Experiments:

Perform any 02 experiment from each group. :

Group A :

1. Introduction to Matlab
2. Operations on Matrix using Matlab
3. Generation of sine, square, triangular, and sawtooth waveforms using Matlab
4. Time Domain Convolution of two signals using Matlab

Group B :

5. Introduction to Multisim
6. RC, RL, RLC Transient Analysis using Multisim
7. Common Emitter AMPLIFIER design using Multisim
8. Op amp applications using Multisim

Group C :

9. Introduction to Proteus
10. Basic Microcontroller Applications such as LED blinking
11. Sensor Interfacing

Group D :

12. Introduction to PCB software
13. Schematic and PCB layout Design
14. To study PCB Etching Process

Text Books :
1. H. Moore, "MATLAB for Engineers", Pearson, 2018. 2. H. S. Kalsi, "Electronic Instrumentation", McGraw Hill, 2010. 3. D. Báez-López and F. Guerrero-Castro, "Circuit Analysis with Multisim", Wiley, 2016. 4. C. Schroeder, "PCB Design for Real-World Design Engineers", Springer International Publishing AG, 2011.
Reference Books:
1. R. L. Boylestad, "Introductory Circuit Analysis", Pearson, 2016. 2. S. R. Gundala, "Microcontroller Programming and Interfacing Using Proteus", Laxmi Publications, 2019. 3. B. R. Hunt, R. Lipsman, and J. M. Rosenberg, "A Guide to MATLAB", 6th ed., Cambridge University Press, 2014..
E-Resources:
1. Multisim Download - NI 2. PCB Tutorial Videos - Learn how to use Proteus EDA Tools 3. MATLAB Onramp (Free introductory course) https://matlabacademy.mathworks.com/

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Program: B. Tech. (Electronics and Computer Engineering)								**Semester:** III	
Course: Internship – II								**Code:** ECIN302	
Teaching Scheme (Hrs/week)				**Evaluation Scheme (Marks)**					
Lecture	**Practical**	**Tutorial**	**Credit**	**CIE**	**ETE**	**TW**	**OR**	**PR**	**Total**
-	-	-	02	-	-	25	-	-	25

Preamble:

Internships serve as vital educational and career development experiences, offering practical exposure in a specific field. Employers seek individuals who possess the necessary skills and an understanding of industry environments, practices, and cultures. This internship is designed as a structured, short-term, supervised training program, often centered on specific tasks or projects with clear timelines. The primary goal is to immerse technical students in an industrial setting, providing experiences that cannot be replicated in the classroom. This exposure aims to develop competent professionals who understand the social, economic, and administrative factors influencing the operations of industrial organizations.

Course Objectives:

1. Exposure to students to the industrial environment, which cannot be provided in the classroom and hence creating deployable professionals for the industry.
2. Learn to implement the technical knowledge in real industrial situations.

Course Outcomes: After completion of this course, students will be able to -

CO1	Gain exposure to industry practices and understand how academic concepts are applied in professional settings.
CO2	Develop and demonstrate effective communication and teamwork skills within a work environment.
CO3	Improve your problem-solving and time management skills by working in real-world industry settings.

Internship Requirements

1. **Internship Duration:** It is mandatory for all students to undergo an internship after every semester during vacations for the duration of 4 weeks. Internships completed during this period will be considered for the assessment of Term Work (TW).
2. **Internship Opportunities:** Students can explore various opportunities for internships at:
   a. Industries
   b. Research labs or organizations
   c. Collegiate clubs
   d. In-house research projects
   e. Online internships
3. **Support and Assistance:** Students can seek assistance for securing internships from:
   a. The Training and Placement cell, along with departmental coordinators
   b. Department or institute faculty members
   c. Personal contacts
   d. Directly connecting with industries or organizations

4. **Request Letter:** Once an industry, research organization, or collegiate club is identified, students must obtain a request letter from the concerned department or placement office. This letter, in the standard format must be duly signed by the authority, should be addressed to the HR manager or relevant authority.
5. **Confirmation Letter:** Students must submit the confirmation letter from the industry, research organization, or collegiate club to the Internship Coordinator and the Head of Department (HOD) office.
6. **Joining Report:** Upon commencing the internship, students must submit the joining report, joining letter, or a copy of the confirmation email to the Internship Coordinator and the HOD office.
7. **Faculty Mentor:** A faculty member will be assigned as a mentor to a group of students. The mentor will be responsible for monitoring, evaluating, and assessing student internship activities. The faculty mentor is also required to visit the internship location and submit formal feedback to the Internship Coordinator.
8. **Faculty Visits:** Faculty members are advised to visit the internship site once or twice during the internship period to monitor progress.
9. **Progress Report:** Students must submit progress report fortnightly to their faculty guide and the final internship report to the Internship Coordinator and department office.
10. **Evaluation Report:** After the completion of the internship, the mentor, along with the assessment panel members, should submit the evaluation report of the students to the department office and the Internship Coordinator.
11. **Internship Certificate:** Students must receive the Internship Certificate from the industry and submit it to the Internship Coordinator and department office.
12. **Presentation and Assessment:** Students are required to give a presentation on their internship work as part of the term work. The internship diary and report will also be verified and assessed.